NG5001 Academic Communication for Graduate Researchers

Unit Two

Writing Academically

First Semester

2025-2026

Learning Outcomes

This unit focuses on writing conventions that different academic disciplines share, which facilitate communication in the wider academic community. Students will explore how these essential features engage and guide readers, as well as improve clarity and flow in academic writing. Students will then apply what they have learnt to write a conference proposal abstract for a cross-disciplinary conference.

Students will:

- use appropriate academic writing strategies to write critically, and to communicate effectively, particularly in cross-discipline contexts
- organize materials for academic communication purposes.

To achieve these objectives, we will explore:

- how metadiscourse operates in academic texts
- how writers create coherence in their writing
- how various writing conventions and strategies contribute to style and voice
- how to use appropriate strategies strategically to write a conference abstract

2.1 ACADEMIC WRITING CONVENTIONS I
(Independent e-learning)

Academic writers often assume that they have to produce a particular style of prose because peer-reviewers and editors will accept nothing else: the kind of impersonal, impenetrable prose that is as stylistically weary, stale and flat as a black academic gown with matching mortarboard. However, more recent evidence has suggested a need to interrogate some of these assumptions.

In this unit, you will debunk some myths about academic writing and be introduced to some features of academic style that are observed in the most effective academic writers across the disciplines. You will recapitulate the general features that make academic writing ‘academic’, and build on that understanding by learning the concepts of rhetorical choices, communication, metadiscourse, and coherence.

As we saw in Unit 1, critical writers not only consider the quality of their evidence and warranting, but also identify how best they can weave these elements into the argument they are developing. They make deliberate choices about the words they use, their sentence style, and their use of paragraphs. They do not simply string ideas together, they interpret evidence for readers, elaborate on their significance, and demonstrate explicitly how ideas relate to each other. In these deliberate choices, they demonstrate care for their readers and also develop their own **voice** (the impression of authorial presence and autonomy of thought).

Voice

Bowden (1999) explains:
"as a metaphor [voice] has to do with feeling-hearing-sensing a person behind the written words, even if that person is just a persona created for a particular text or a certain reading."

Voice is understood as representing a writer's identity, visibility, points of view, and stance in the text (Hyland & Sancho-Guinda, 2012; Tardy, 2016). Put differently, it is like an "accent" that will permeate your writing.

Hyland (2008) proposes that voice is constituted by "the ways writers present themselves and convey their judgements, opinions, and commitments" (p. 7). It is strategically "*crafted, constructed, built, carved, created*" (Sancho-Guinda & Hyland, 2012, p. 5) depending on the academic persona the writer wishes to present to readers.

Research shows that scholars use various metadiscursive, rhetorical, and linguistic devices (incl. word choices and sequencing) to adopt various **personas** or **roles**, depending on the purpose and genre of the text, and often informed by disciplinary norms.
Examples of personas or roles:

- The arguer or interpreter
    - To present claims and analysis, writers foreground their critical stance and reasoning (Bondi, 2012)
- The observer or reporter
    - The knowledge rather than the reasoning is foregrounded (Silver, 2012)
- The self as academic expert
    - Strategic self-representation involving thematic and linguistic choices that imply credibility and authority (Tse, 2012)

Voice also serves to **engage and persuade** the readers, guiding them to align with the writer's arguments or perspectives.

Constructing a strong authorial voice is essential in academic writing. The development of an authorial voice has been granted increased importance in research on academic research writing (see Matsuda & Tardy, 2007; Morton & Storch, 2019, for example).

A thesis or dissertation is a genre expected to convey the writer's voice. However, due to the opaque conventions of academic discourse and the high stakes involved in a dissertation, students may find this daunting.

Wellington et al. (2005, p. 84) offer suggestions for finding your academic voice:

- **authorial presence & autonomy of thought (original ideas and perspectives, without passively accepting or adopting the perspectives of others).**
- healthy scepticism, but not cynicism
- confidence and assertiveness, without 'cockiness' or arrogance
- judgement which is critical but not dismissive
- opinions, without being opinionated
- careful evaluation of published work "not serial shooting at random targets"
- being 'fair': assessing fairly the strengths and weaknesses of other people's ideas and writing without prejudice
- making judgements on the basis of considerable thought and all the available evidence as opposed to assertions without reason.

Metadiscourse provides writers with a resource to accomplish these goals.

Metadiscourse is a concept that refers to the linguistic devices writers employ to assist readers to connect, organise, and interpret material in a way preferred by the writer and with regard to the understandings and values of a particular disciplinary community. Through metadiscourse the writer makes choices to express a 'voice' that is distinctly theirs while consistent with disciplinary norms by revealing a suitable relationship to his or her data, arguments, and audience. The table below provides a summary and classification of the linguistic resources of metadiscourse.

Category	Function	Examples
Interactive resources	Help to guide reader through the text	
Transitions	Express semantic relation between main clauses	In addition/but/thus/and
Frame markers	Refer to discourse acts, sequences, or text stages	Finally/to conclude/my purpose is to
Endophoric markers	Refer to information in other parts of the text	Noted above/see Fig./in Section 2
Evidentials	Refer to source of information from other texts	According to X/(Y, 1990)/Z states
Code glosses	Help readers grasp meanings of ideational material	Namely/e.g./such as/in other words
Interactional resources	Involve the reader in the argument	
Hedges	Withhold writer's full commitment to proposition	Might/perhaps/possible/about
Boosters	Emphasise force or writer's certainty in proposition	In fact definitely/it is clear that
Attitude markers	Express writer's attitude proposition	Unfortunately/I to agree/surprisingly
Engagement markers	Explicitly refer to or build relationship with reader	Consider/note that/you can see that
Self-mentions	Explicit reference to author(s)	I/we/my/our

Metadiscursive resources are essential to good writing because they cue readers to interpret texts in ways desired by authors, and make the writing more explicit and more reader-friendly. In other words, they help to create coherence in a text.

Coherence is commonly described as the relationships that link the ideas in a text to create meaning for the readers. These connections are not limited to the micro level of the text (e.g. words and sentences), but extend to the macro level (e.g., paragraphs, sections, chapters, etc.).

More specifically, coherence can be defined in terms of the following features:

1. A **macrostructure** with a characteristic pattern or shape appropriate to its communicative purpose and context;

2. An **information structure** that guides the reader in understanding how information is organized and how the topic of text is developed (simply put, coherent texts often comply with the principle of giving old information before new information);
3. Connectivity of the underlying content evidenced by **relations between propositions**;
4. Connectivity of the surface text evidenced by the presence of **cohesive devices**; and
5. Appropriate use of **metadiscourse**.

Understanding how these elements of coherence work in texts will help us to use them appropriately to develop coherence in our writing.

Task 1: Reading and Viewing

(a) View E-Lecture 3A and E-Lecture 3B (Canvas Home > Unit 2) to learn the concepts of metadiscourse and coherence that are essential to effective academic writing.

(b) *Optional*:
Read Hyland (2004), focusing on section 1 (the concept of metadiscourse) and section 3 (an interpersonal model of metadiscourse), to deepen your understanding of metadiscourse and its relevance to graduate student writing.

Task 2: Online Discussion

Study the following extract from the introduction of a journal article.

Connections

Apply what you have learnt about metadiscourse in this Unit. Post responses to your group's forum in Canvas Discussions, answering the following questions:

a) Identify one or two instances of metadiscourse that demonstrate skillful choices by the authors. Explain briefly how these words and phrases show care for readers, making it easy for readers to follow the ideas in the text.

b) What do you think the authors' linguistic choices reveal about their stance on the topic and their relationship with the existing research? Look at 1-2 introductions from peer-reviewed journal articles in your discipline. Pay attention to the metadiscourse patterns (e.g. how authors use metadiscourse to guide readers, express stance). Do you observe any significant differences in how metadiscourse is used and how authorial voice is constructed in your field compared to the example provided?

Your answer to the questions above should be about 100-200 words each.

A good night's sleep is crucial for the physical and mental wellbeing of a person. However, poor sleep is a growing public health concern. Multiple studies have reported a significant prevalence of poor sleep quality and insomnia across Asia [[1], [2], [3], [4]]. Using the general cut-off of PSQI score of >5, a total of 26.4% (for men) to 31.1% (for women) of a Japanese sample reported poor sleep [2], about 27% of people from China reported insomnia [3,4] and 39.4% of a region-representative population in Hong Kong reported poor sleep [5]. In Singapore, the Singapore Health 2012, a population wide study reported that 27.2% of respondents had poor sleep [6].

Poor sleep has been linked with poorer health outcomes in a number of studies [[7], [8], [9]]. Notably, insomnia has been linked with metabolic syndrome [8], multimorbidity [7] and mental health issues [[9], [10], [11]]. Besides health outcomes, poor sleep has been associated with unhealthy lifestyle habits like smoking [3] and excessive alcohol use [12].

Due to the importance of sleep in physical and mental wellbeing, significant attention has focused on the poor sleep patterns and insomnia among the public. For example, studies have shown that women [13], persons of minority ethnic groups [14,15], persons with obesity [16], and persons of lower socioeconomic status [17] were generally more likely to report poorer sleep patterns.

Besides the Singapore Health 2012 survey [6], there is a paucity of epidemiological studies looking at nationwide prevalence of poor sleep quality and insomnia in Asian countries. Most Singaporean reports on sleep quality focused on specific demographic groups, like working adults [18,19] or elderly patients [20]. An oft-cited report series commissioned by Philips to track sleep quality around the world (including Singapore) used a non-probability sampling method, which inevitably contains a margin of error that cannot be accounted for [21]. Such non-representative or convenience samples tend to over-estimate the true prevalence of insomnia due to respondent bias; respondents with sleep issues are more likely to take part in a survey on sleep quality [22]. Moreover, sleep quality is associated with a myriad of sociodemographic factors. Existing Singapore reports based on non-representative samples like the ones of specific demographic [[18], [19], [20]] do not capture the true prevalence of poor sleep in the Singapore population.

Even though polysomnography is considered the gold standard in diagnosing sleep disorders in patients [23], it is not feasible to recruit a large number of participants often amounting to thousands in the general population for polysomnography data. Instead, PSQI is the most used self-reported scale as a proxy to obtain information on sleep quality from the respondents [24]. PSQI delves into seven aspects of sleep, including sleep pathology, subjective sleep quality, latency, duration, habitual sleep efficiency, sleep disturbances, use of sleep medications, and daytime functioning. For the sake of brevity, these seven components were taken together as a single score to provide a proxy of the respondent's sleep “quality”.

The aims of this study were to 1) characterize the sleep quality of a nationally representative sample of Singapore residents using PSQI and 2) identify the sociodemographic correlates of poor sleep in this population. As sleep quality is different

from sleep quantity, we have segregated the data analysis and presentation into two parts. One deals with the sleep quality of this sample population, while the other focusses on sleep quantity. With this information, clinicians could identify people who are more vulnerable to sleep issues and deliver targeted psychoeducation. These data may also help to provide a more accurate view on the status of Singapore residents' sleep quality.

2.2 WRITING THE CONFERENCE PROPOSAL ABSTRACT I (Synchronous Tutorial)

In the e-learning materials, we have learnt some concepts that are essential to effective academic writing. These concepts include choices, communication, metadiscourse, and coherence and voice. In this tutorial, we will deepen our understanding of some of these concepts to help us write our abstract

Task 3: Viewing & Drafting

Please complete before attending tutorial

(a) View E-Lecture 4A and E-Lecture 4B (Canvas Home > Unit 2) to learn how to write a conference abstract.
(b) Start drafting your own basic abstract. Please be reminded that you will need to hand in your draft by the end of this week (Week 4).

Optional

To help you, you may want to read Pallas (2022), Chapter 7 "Applying to your First Conference and Writing an Abstract".

Task 4: Voice

Read the excerpt below, taken from a paper from microbial ecology.

Q. Can you identify the language choices that contribute to the writer's voice? Consider the selection of words and phrases. What impression of the writer or researcher does this language create?
Q. How did the author's voice (authorial presence) influence the connection with readers and the persuasiveness of the writing?

> Larry Pomeroy's seminal paper revolutionized our concepts of the ocean's food web by proposing that microorganisms mediate a large fraction of the energy flow in pelagic marine ecosystems. Before 1974, bacteria and protozoa were not included as significant components of food web models. Pomeroy argued forcefully that heterotrophic microorganisms, the "unseen strands in the ocean's food web," must be incorporated into ecosystem models.
>
> From:
> F. Azam, D. C. Smith, G. F. Steward, and A. Hagström, "Bacteria-Organic Matter Coupling and Its Significance for Oceanic Carbon Cycling," *Microbial Ecology* 28 (1994): 167–79.

Task 5: Using Metadiscourse to Improve Flow and Develop Authorial Voice

Revise the following to make them more effective openings. Use metadiscourse to guide the reader through the text improving flow, and adjust the language to project a strong authorial voice.

Keep in mind the following

- Q. who is your audience?
- Q. how do they need to hear this & therefore what persona* will you project

1. The rates of all chemical reactions increase with temperature.
   This phenomenon grows directly from physical chemistry's transition state theory and the Arrhenius equation. Respiration in soil doesn't always appear to follow this pattern. Some studies have shown no respiration response to increasing temperature, while a few have even reported a negative response.

2. Chemotherapy is a dominant treatment approach for many types of cancer, and with the development of new targeted-delivery systems has the potential to become even more widespread and efficacious. A common constraint to effective chemotherapy is patient resistance to the treatments. Such resistance is often closely associated with the activity of the enzyme γ-glutamyl transpeptidase (GGT), which acts to increase intracellular concentrations of glutathione and thereby block the apoptotic cascade in tumor cells. Inhibiting GGT before chemotherapy would therefore reduce tumor cell resistance and increase treatment effectiveness.

Follow up activity:

- Revise the text using ChatGPT to establish your intended authorial voice.
- Compare your revision with the one generated by ChatGPT. Reflect on the strengths and limitations of using ChatGPT for writing.

Considering Style

As you develop your abstract into a presentation, and eventually into a full paper, or dissertation, you will need to start thinking more about your writing style.

Attention to academic style is often associated with editing and the final stages of writing, but it is never too early to develop an awareness of style. Writers need to develop the skills necessary for communicating within stylistic conventions even when writing early drafts. This means balancing complexity and clarity, attending to the stylistic norms of the discipline(s), while maintaining awareness of context. In this section, we will consider in more detail what a suitable academic style looks like, across a number of disciplines.

The following is advice given to students:

When you struggle to understand some academic writing (and you will), don't blame yourself, at least at first. Diagnose its sentences. If they have long subjects stuffed with abstract nouns, expressing new information, the problem is probably not your inability to read easily, but the writer's inability to write clearly (Turabian et al., 2023, p. 137).

This quote reflects the increasing tendency for writers to "self-promote" through the excessive use of jargon and flowery rhetoric, as academic life becomes increasingly competitive. Writers think that this style of writing will make their voice heard more loudly (Thomas, 2016, p. 133).

This raises the question of how much rhetoric and jargon is too much. There is no simple answer. One must first look to the conventions of the discipline and also at the types of writing accepted in journals you wish to publish in**.**

In cross-disciplinary contexts, when advocating in the wider academic community, there is the added issue of ensuring you do not alienate readers/ listeners from other disciplines.

While you may be inspired to imitate the distinctive style of notable scholars, you will need to exercise some caution here. For a student writing and presenting a graduate-level thesis, rhetorical excess is not realistic; clarity and reader expectations must be priorities.

This is not to say that you must write in a highly formulaic or restrained manner, especially if it does not feel natural to you. After all, even the most well-known scientists used literary rhetoric (e.g. Darwin, Oppenheimer). However, you must take care that you are offering a clearly articulated argument and not assume that your writing is understood by all of your readers/ listeners. Always make sure that you are not simply trying to camouflage the lack of meaning or significance with rhetorical excess.

This checklist may be useful when writing and revising.

Avoid:

- unnecessarily long or complex sentences
- unnecessarily short disconnected sentences
- overuse of abstract nouns
- complex structures where grammar and meaning do not match
- repetition (unintentional and/or with no clear purpose)
- redundancy
- gratuitous jargon
- vagueness or ambiguity
- excessive imagery or obscurity

(Thomas, 2016, p. 121)

Introducing the conference proposal abstract

Abstracts are often mentioned but rarely discussed and can therefore be something of a mystery to conference newcomers. Conference abstracts are relatively short texts that are generally submitted before the conference as part of the selection process and which then go on to serve multiple purposes during and beyond the lifetime of the conference. Typically, abstracts include the title of the presentation and between 200 and 500 words on what will be covered. Sometimes an abstract is the only text which is required when proposing a presentation to a conference. In this case, the abstract is particularly important as it is the only basis for the conference organisers to decide who will present at the conference and who may not be accepted to present. Though the abstract is only a short text, its importance is therefore rather inflated— and crafting a 'good' abstract is an art in itself.

A conference abstract is charged with different tasks. It needs to meet the requirements of various audiences. For instance, the abstract needs to meet the expectations of the conference organisers. This may include demonstrating relevance to the theme and/or disciplinary orientation of the conference; adhering to any requirements in the call for papers (e.g., length, style, elements to include); and showing evidence of the competence and expertise of the author. A diligently produced abstract may meet all the

organisers' needs but have zero appeal for other conference attendees. The abstract, and indeed the title of the presentation, therefore need to include some appeal—why would attendees choose to come along to this paper, rather than the others running at the same time—or rather than skipping the session to have a chat or catch up on email? The appeal needs to be balanced with the organisers' requirements—too much excitement and the organisers may not take the abstract seriously. Much of this is about tuning into the vibe of the conference, which fundamentally begins from reading the information in the call for papers carefully.

The conference abstract is not an abstract for the project as a whole—this is a mistake often made by novice abstract writers. It is often not possible to present a whole project in 15–20 minutes, and the abstract needs to indicate the specific focus of the presentation. An abstract therefore needs to indicate that the presentation is drawing on a wider project, with some brief information about this, but the presentation—its topic, structure, argument—is the focus of the abstract.

More than just a summary, an abstract is a microcosm of your work that allows the reader to assess, in just a minute or two, whether you are asking an interesting question, using robust methods, and have interesting findings. Put another way, an abstract makes an implicit argument for why your work is important and needs to be presented publicly

Belcher (2021) on abstracts:

ABSTRACTS AS A TOOL FOR SUCCESS

'One of the best early tasks you can undertake to improve your journal article is to write an abstract—something that describes your article's topic and argument. In fact, one scholar published an article about how she **dramatically improved her writing productivity after learning…"to write the abstract first,"** as the task became "the anchor and catalyst for the framing and reframing of writing goals" (Linder et al. 2014, 223). Unfortunately, many scholars see writing an abstract as the last step to publication. In the humanities, they may never have to write one! But writing an abstract, regardless of whether journals in your field require one, is an important step in revising your article, not a mere formality. More than one authority has noted that "a well-prepared abstract can be the most important paragraph in your article" (APA 2010, 26)'

(Belcher, pp.92)

How to write a conference proposal abstract
The main components of a good abstract are illustrated as follows:



Note:
The above is the structure of abstracts written in the scientific and quantitative disciplines (SciQua). Abstracts from the **humanities and interpretive disciplines** (HumInt) have less in common and are not as prescriptive, however, good HumInt abstracts generally include:

- Background
- Some evidence
- An indication of the theoretical framework(s)/ approach
- Claim for significance
- The argument

(Belcher, 2021)

This basic structure also works for a **work-in-progress**, although the discussion of methods, findings, and significance may look slightly different. Pallas (2022:122) gives a useful example of how the basic abstract structure may be modified for works-in-progress. Instead of drawing on empirical data, the methods may, for example, develop a theoretical framework that is used to analyze the literature. The findings then may be the results of that analysis, and the significance would likely be to justify a future research agenda. The research question may also need to be revised since the goal of a work-in-progress is likely to set up empirical research rather than report on it.

Some helpful sentence structures that can be used to produce a basic abstract are:

1. "Existing research on [topic] argues/claims..."
2. "However, [new data/theory/analysis] indicates/reveals..."
3. "To address this problem/limitation/debate, this paper asks: [research question]"
4. "This paper answers this question using a [research design or data source]" OR "This paper answers this question using an analysis of the literature on..."
5. "These [data/analyses] indicate...[findings]"
6. "These findings suggest that literature on [topic] must [change/adapt/reconsider]..." OR "These findings indicate that the best solutions to [problem] involve..." OR "These findings suggest that future research should..."

Note that the maximum length of the abstract will vary depending on the call for papers. It can be as little as 100 words or as much as 1000; 200–300 words is very common. You may not exceed the stated length, but you also should not be too far under it. A call asking for a 700-word abstract is looking for a high degree of detail; submitting a 200-word version probably will not satisfy the reviewers. Pay attention too to any requests for sources. Abstracts typically do not include citations, but some calls – typically those requesting longer abstracts – require them. If that is the case, read the call for papers carefully to find out if the citations are included in the word count for the abstract.

Wallace and Wray (2021) provide a systematic way of asking yourself focus questions about the conference abstract as you write, and orienting your composing process in a way that addresses the needs and interests of your intended audiences:

Focus question	**Orientation of the text**	**How this addresses reviewers and attendees**
Why might this be of interest to people at the conference?	Make clear early in the text what topic and issues you are addressing (e.g., 'The central question addressed in this paper is…').	• Reviewers need to know that your presentation is relevant to the conference themes. • Attendees need to know why they would find your paper

		relevant and interesting.
Why is this an important question to ask?	Locate your work within the existing research literature just enough to show what sort of previous research is relevant (e.g., 'With particular reference to the theory of X and Y…'); refer only to central works.	• Reviewers are judging whether you are familiar with the context of your work, or just doing something that is isolated and hence not likely to be useful to others. • Attendees are looking for points of contact between your work and theirs, so they know they will understand what you are doing and see the relevance to their own work.
What did I do?	Say what method you used (e.g., 'Interviews were conducted to…'; 'Using a series of experiments to measure…'; 'Texts were gathered from… and analysed for…').	• Reviewers are gauging whether your methods were reliable. They are gatekeepers of research quality and want to ensure you don't mislead your audience. • Attendees want to know if your method is like theirs (e.g., what ideas they might get for their own work) or whether you will be reporting a type of data they believe is interesting and valid. In a presentation, you won't have the time to describe methods in

		the detail necessary for the audience to judge their quality.
What are my main claims?	Give the headline findings (e.g., 'Overall, nurses were more likely than doctors to attribute low performance to under-resourcing, while doctors blamed poor management').	• Reviewers are trying to establish whether your claims seem plausible, given your aims and method. They may also be assessing the novelty and significance of your work. • Attendees are looking for the hook – will this be an interesting presentation? Here you try to sell your work as worth attending because of what it can tell them about something in the world.
How might readers find this useful?	Suggest the potential for generalization beyond your own study to other contexts which might be similar in significant ways (e.g., 'It can be inferred from these results, that all teenagers from ethnic minority groups are…'; 'It remains unclear whether these findings would hold for other populations of care home residents because…').	• Reviewers are trying to judge whether your findings are likely to be relevant to attendees. If they can't find evidence of generalizability, they won't see much point in you presenting the work to others. • Attendees are always looking for that hook from your work into theirs. If your work can't generalize to a domain of their interest they won't bother coming.

Characteristics of a good conference abstract

To determine what conference organizers are looking for in an abstract, we need to go inside their heads. Organizers want an interesting, diverse, and entertaining meeting devoted to the advertised topic. Therefore, if you wish to get invited to the conference, you need to address three things, in order of easiest to most difficult:

1. Your abstract must be relevant to the conference theme.
2. Your abstract must be of interest to the conference audience.
3. Your abstract must convince the committee that you are capable of giving a high-quality presentation.

Consequently, the best abstracts have the following attributes:

- *New*. An abstract that presents new results is more likely to get accepted than the same material you presented at last year's conference.
- *Interesting.* The abstract should discuss the unresolved aspects of the research topic, express your contribution, and suggest that you have an exciting presentation to deliver. Because the desire of most conference committees is to have a lively meeting with lots of dialogue or debate, conference abstracts can be more controversial than the typical article abstract.
- *Focused*. Given the relatively short length of most conference presentations, you generally cannot present an entire manuscript, let alone your entire dissertation. Pick one or two points that you want to make in the abstract and spend time motivating, developing, and demonstrating these points in your presentation.
- *Informative*. Vague abstracts with little scientific substance, particularly results, are not attractive to the conference committee; the committee cannot risk taking the chance that your presentation will be devoid of content. Be specific about your results in the abstract, avoiding vague statements like "results will be discussed" without saying what at least some of those results are. However, if you reveal all your results and conclusions, you take away the motivation for some in the audience to see your presentation, especially those that are only marginally interested in your topic. Therefore, you may prefer a more coy approach by not disclosing all your results in the abstract.
- *Understandable*. Because the audience will generally be much broader (e.g., more varied education, more diversity in research interests) at most conferences than the audience of your journal article, more background information, fewer technical details, and more motivation for why your work is important need to be presented in the conference abstract.
- *Enticingly titled*. The title should accurately describe the work, but also entice people to see it. The title should encapsulate the spirit of the presentation.
- *Well written*. The abstract should make compelling arguments, be well organized, and be free of grammatical errors and typos.

Task 6: Analysing a Sample Abstract

Below is a sample adapted from an abstract written by Ms Krithi Pushpanathan from the NUS Department of Ophthalmology, for the specific purpose of the NG5001 Assignment I. She has very kindly given her consent for us to study her abstract for the same purpose. We are grateful for her generosity.

Examine the sample and answer the following questions:

1. Is the study well-located in the knowledge of the field? In what ways does the writer engage with existing scholarship in order to establish a niche and assert her own stance?
2. Is the research purpose well-justified? Is it compelling, targeted and clear? How does the writer use language to persuade and align the reader with her perspective?
3. What research design, methods or approaches were adopted, and why? How well-aligned are they to the research question?
4. What are the main results?
5. What do the results mean for theory and/or practice? How do they relate to the course conference theme?
6. In what ways does the writer attend to the cross-disciplinary context and audience?
7. In what ways does the writer's language and use of evaluative expressions capture authorial voice?

Follow up activity:

Review your own research purpose. Is it clearly defined, compelling and well-justified? Does your review of relevant scholarship effectively position your work in the field and highlight your distinctive contribution? Do your background, methods, results align with your research question? Are there any areas that could be better tailored to your objectives?

Evaluating Large Language Model Chatbots for Addressing Patients' Ocular Symptom Queries

Abstract:
Large language models (LLMs), notable for replicating human intelligence through rigorous training on diverse datasets, are emerging as valuable tools in medicine.[1] These chatbots generate contextually appropriate responses to user queries.[2] Due to their widespread accessibility, user-friendly interface, and versatility, LLMs are employed for various medical functions including appointment scheduling, medical record documentation, and clinical decision support.[1-3] Furthermore, their competency in answering medical exam questions[4-6] and ability to express empathy[7-9] have fostered pervasive user willingness for self-diagnosis[10].

While LLMs hold promise for aiding in patient triage, their propensity to disseminate misinformation poses a significant challenge to consistently conveying accurate medical guidance to naïve patients.[11-13] For text-based responses, LLMs predict subsequent

words from context, thus possibly providing convincing yet erroneous responses and threating patient safety.[14] This concern is especially pronounced in specialties requiring nuanced clinical reasoning, such as ophthalmology, where errors in symptom interpretation can lead to potentially serious consequences, such as delayed or inappropriate care. In addition to concerns about accuracy, scholars have raised issues about LLMs' transparency, explainability, and their susceptibility to prompt engineering or adversarial inputs, which can further distort output.[16] Beyond issues of accuracy, recent studies suggest that LLMs perform inconsistently across medical contexts. ChatGPT's accuracy was found to be comparable to that of untrained doctors in emergency care, emphasizing the need for domain-specific evaluation prior to clinical application.[17] Taken together, these findings reinforce the need to assess LLM behaviour in speciality-specific scenarios, such as opthamology, where accurate initial assessment is critical.

Limiting the scope to triaging ophthalmic symptoms, only Tsui et al (2023) had explored the proficiency of LLMs in this domain but utilising only one LLM, specifically ChatGPT-3.5, and on a small set of 10 questions.[15] Our study expands on this by assessing the accuracy and comprehensiveness of responses from ChatGPT-3.5, ChatGPT-4.0, and Google Bard, to a wider array of 37 commonly-inquired patient concerns regarding ocular symptoms. This approach addresses a gap in the literature by comparing the performance of multiple LLM versions and platforms, providing a more comprehensive understanding of their clinical reliability in eye care.

We shortlisted common ophthalmology symptom inquiries from reputable online health sources like the National Eye Institute and the American Academy of Ophthalmology. Three consultant-level ophthalmologists polished them based on clinical experience to create our final question set. Chatbot responses were shuffled and graded independently by the ophthalmologists in a blinded manner over separate days with washout periods to prevent carry-over effects. Accuracy was graded on a 3-point scale (poor, borderline, good) through consensus. 'Good'-rated responses were further evaluated for comprehensiveness on a 5-point scale. For 'poor'-rated responses, the chatbot was prompted to self-correct, and revised responses were re-graded.

Overall, ChatGPT-4.0 (8.19) scored significantly higher than ChatGPT-3.5 (7.46) and Google Bard (7.00) for accuracy, out of a maximum of 9 ($p<0.001$). While ChatGPT-4.0 produced the highest proportion of 'good'-rated responses (89.2%), it still generated 10.8% 'borderline' responses, indicating room for improvement. On the other hand, comprehensiveness was comparable across all three chatbots. Self-correcting abilities in ChatGPT-3.5 and Google Bard were moderate, improving 6 of 7 'poor'-rated answers, but predominantly resulting in 'borderline' grades only.

Our study demonstrates the potential role of LLMs, particularly ChatGPT-4.0, in alleviating healthcare burdens by assisting in triaging of ophthalmic symptoms remotely, in real-time and at low or no cost, thereby improving patient access to care.

References

1. Singhal, K., Azizi, S., Tu, T., Mahdavi, S.S., Wei, J., Chung, H.W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., et al. (2023). Large language models encode clinical knowledge. Nature. 10.1038/s41586-023-06291-2.
2. Clusmann, J., Kolbinger, F.R., Muti, H.S., Carrero, Z.I., Eckardt, J.-N., Laleh, N.G., Löffler, C.M.L., Schwarzkopf, S.-C., Unger, M., Veldhuizen, G.P., et al. (2023). The future landscape of large language models in medicine. Communications Medicine *3*. 10.1038/s43856-02300370-1.
3. Alowais, S.A., Alghamdi, S.S., Alsuhebany, N., Alqahtani, T., Alshaya, A.I., Almohareb, S.N., Aldairem, A., Alrashed, M., Bin Saleh, K., Badreldin, H.A., et al. (2023). Revolutionizing healthcare: the role of artificial intelligence in clinical practice. BMC Medical Education *23*. 10.1186/s12909-023-04698-z.
4. Kung, T.H., Cheatham, M., Medenilla, A., Sillos, C., De Leon, L., Elepaño, C., Madriaga, M., Aggabao, R., Diaz-Candido, G., Maningo, J., and Tseng, V. (2023). Performance of ChatGPT on USMLE: Potential for AI-assisted medical education using large language models. PLOS Digit Health *2*, e0000198. 10.1371/journal.pdig.0000198.
5. Cai, L.Z., Shaheen, A., Jin, A., Fukui, R., Yi, J.S., Yannuzzi, N., and Alabiad, C. (2023). Performance of Generative Large Language Models on Ophthalmology Board Style Questions. Am J Ophthalmol. 10.1016/j.ajo.2023.05.024.
6. Schubert, M.C., Wick, W., and Venkataramani, V. (2023). Performance of Large Language Models on a Neurology Board–Style Examination. JAMA Network Open *6*, e2346721. 10.1001/jamanetworkopen.2023.46721.
7. Bernstein, I.A., Zhang, Y.V., Govil, D., Majid, I., Chang, R.T., Sun, Y., Shue, A., Chou, J.C., Schehlein, E., Christopher, K.L., et al. (2023). Comparison of Ophthalmologist and Large Language Model Chatbot Responses to Online Patient Eye Care Questions. JAMA Network Open *6*, e2330320. 10.1001/jamanetworkopen.2023.30320.
8. Brin, D., Sorin, V., Vaid, A., Soroush, A., Glicksberg, B.S., Charney, A.W., Nadkarni, G., and Klang, E. (2023). Comparing ChatGPT and GPT-4 performance in USMLE soft skill assessments. Scientific Reports *13*. 10.1038/s41598-023-43436-9.
9. Ayers, J.W., Poliak, A., Dredze, M., Leas, E.C., Zhu, Z., Kelley, J.B., Faix, D.J., Goodman, A.M., Longhurst, C.A., Hogarth, M., and Smith, D.M. (2023). Comparing Physician and Artificial Intelligence Chatbot Responses to Patient Questions Posted to a Public Social Media Forum. JAMA Internal Medicine. 10.1001/jamainternmed.2023.1838.
10. Shahsavar, Y., and Choudhury, A. (2023). User Intentions to Use ChatGPT for SelfDiagnosis and Health-Related Purposes: Cross-sectional Survey Study. JMIR Human Factors *10*, e47564. 10.2196/47564.
11. Azamfirei, R., Kudchadkar, S.R., and Fackler, J. (2023). Large language models and the perils of their hallucinations. Critical Care *27*. 10.1186/s13054-023-04393-x.
12. Stokel-Walker, C., and Van Noorden, R. (2023). What ChatGPT and generative AI mean for science. Nature *614*, 214-216. 10.1038/d41586-023-00340-6.
13. van Dis, E.A.M., Bollen, J., Zuidema, W., van Rooij, R., and Bockting, C.L. (2023). ChatGPT: five priorities for research. In Nature, pp. 224-226. 10.1038/d41586-023-00288-7.
14. De Angelis, L., Baglivo, F., Arzilli, G., Privitera, G.P., Ferragina, P., Tozzi, A.E., and Rizzo, C. (2023). ChatGPT and the rise of large language models: the new AI-driven

infodemic threat in public health. Frontiers in Public Health *11*. 10.3389/fpubh.2023.1166120.
15. Tsui, J.C., Wong, M.B., Kim, B.J., Maguire, A.M., Scoles, D., Vanderbeek, B.L., and Brucker, A.J. (2023). Appropriateness of ophthalmic symptoms triage by a popular online artificial intelligence chatbot. Eye. 10.1038/s41433-023-02556-2.
16. Pushpanathan, K., Lim, Z.W., Er Yew, S.M., Chen, D.Z., Hui'En Lin, H.A., Lin Goh, J.H., Wong, W.M., Wang, X., Jin Tan, M.C., Chang Koh, V.T., and Tham, Y.-C. (2023). Popular large language model chatbots' accuracy, comprehensiveness, and self-awareness in answering ocular symptom queries. iScience *26*, 108163. 10.1016/j.isci.2023.108163.
17. Masanneck, L., *et al* (2024). Triage performance across large language models, ChatGPT, and untrained doctors in emergency medicine: comparative study. *JMIR Medical Informatics, 26*, e53297

Task 7: Conference Abstract DRAFT

Study carefully the requirements of Assignment I as described in the Course Handbook. Compose (write) a draft for this assignment and upload your draft to Canvas Assignments (Conference Abstract - Draft). The deadline for uploading your abstract is: **Friday 5 September, 17:00 (5pm)**

You should try to make your draft as complete as you can. Your tutor will go through your draft during consultation sessions in weeks 5 & 6.

Establishing Common Ground

Disciplinary differences – in research conventions, methodologies, epistemologies, parameters of legitimacy—represent significant barriers to communication. In addition to this, various terms have different meanings in different disciplines.

The real challenge in communication is not simply exchanging knowledge, which is relatively straightforward. It's about finding ways to express the tacit knowledge that comes from years of immersion in the unique practices of a particular field.

Just as good conversations require speakers to share a common language or means for understanding one another, disciplines need a way to communicate to one another that makes sense and crosses linguistic boundaries.

Finding "common ground" involves finding key words, concepts, or mental models that are understood across disciplines. In this sense, common ground is a bridge of communication between disciplines.

There will always be a knowledge gap between you and your audience, even if you are from the same discipline, as scholarship even within disciplines is quite diverse. If your audience represents multiple disciplines, the distance between you and your listener is even bigger.

Establishing common ground is an essentially **creative act** that attempts to connect with the audience, and bridge the knowledge gap, so that effective communication can take place. To do this, a scholar must be open minded and aware of other, non-disciplinary perspectives, and aware of the assumptions of one's own discipline, particularly in relation to communication.

Establishing a common ground in communication across disciplines can include:

- Using shared mental models (e.g. using examples, shared experiences, metaphors, familiar verbal or graphical analogies that are relatable to various audiences to describe complex ideas)
- Using inclusive language, and avoiding jargon or technical terms that might alienate
- Using a non-specialist academic style (discussed in Unit 2.1)
- Developing a strong narrative to engage audiences, or humanizing your research (this is especially powerful in oral presentations and longer pieces of academic writing; see also Unit 3).

Task 8: Common Ground

Read the following abstracts.
How have the authors established common ground?

Text A

Trust is a complex phenomenon that resists easy definition, but it is easily recognizable, or rather its absence is impossible to miss. The author draws inspiration from the #MedsWeCanTrust movement to advocate for #MedicineWeCanTrust. Trust can be seen as a "soft," "tender-minded," optimistic condition fighting for survival in a "hard," "tough-minded," or jaundiced medicine. Modern medicine is traditionally patriarchal, individualistic, and resistant to encouraging democratic, collaborative habits as it socializes its young into hierarchical structures or eats them whole. Yet trust is a health intervention and essential for the innovative expansion of medical culture as it encourages authentic democracy, interprofessional clinical teamwork, and patient-centeredness. Increases in trust lead to greater tolerance of uncertainty, one of the primary goals of medical education. Recent curriculum development work has shown that the medical humanities offer a superb delivery mechanism for ensuring democratic habits in medicine that align with social justice agendas, key to addressing links between social inequalities and compromised physical and mental health. Where lack of trust is associated with cynicism in doctors, increasing trust loosens dependence upon suffocating control mechanisms. This allows medicine to take on the moral concerns and uncertainties of an adulthood that also promises emotional warmth, guidance, support, and improved communication between colleagues and with patients. Medicine must embrace trust as the matrix of health care, and the medical humanities can educate for values such as tolerance of uncertainty and ambiguity as a basis for engendering trust.

Bleakley A. (2019). Invoking the Medical Humanities to Develop a #MedicineWeCanTrust. *Academic medicine : journal of the Association of American Medical Colleges*, *94*(10), 1422–1424.

Text B

In this study, we looked at how to treat patients with a severe form of ulcerative colitis, a type of inflammatory bowel disease, when the usual treatments don't work. We tested a combination of two drugs, cyclosporine and ustekinumab, to see if it could help these patients in the long term. We included eleven patients who had already tried many other treatments and didn't get better. Most of them had also tried a drug called infliximab and had failed at least three other biological therapies. At the start, these patients were very sick, with high scores on disease activity measures. We gave them ustekinumab in addtion after a therapy with cyclosporin had been started before. The combination therapy continued for an average of almost 12 weeks. After 16 weeks, six patients showed improvement in their symptoms, and five were able to stop taking steroids. Five patients also had their colon lining looking healthy again when we looked inside with a scope after 16 weeks. And after 52 weeks, five patients had normal colon lining and healthy tissue under the microscope. Ultrasound showed that the thickness of their colon wall had decreased. Unfortunately, two patients had to have surgery to remove their colon, and three had to stop taking ustekinumab because it didn't help them. Overall, the combination therapy was safe and well-tolerated. In conclusion, combining cyclosporin and ustekinumab helped about half of the patients with severe ulcerative colitis get better and have healthy colon lining after 52 weeks. This suggests that more research is needed to understand the benefits of this treatment in these patients.

Vitali F, Rath T, Klenske E, et al. Long-term outcomes of cyclosporin induction and ustekinumab maintenance combination therapy in patients with steroid-refractory acute severe ulcerative colitis. *Therapeutic Advances in Gastroenterology*.

Text C

When medical studies have collected large amounts of health data on people classified in the data as having a particular disease, it may often be difficult to "see the wood from the trees" in terms of being sure that that disease classification is actually correct for each person in the dataset. Therefore, it is important that researchers first verify the disease classification accuracy of the data against an external "gold standard" source of information holding the correct diagnosis. This is usually a patient's medical records. A comparison of the health data disease classification against the medical records diagnosis generates various measures of disease classification accuracy, all of which have advantages and disadvantages as verification measures. Here, we examine a verification measure which has recently been characterised as "F*". It turns out, in fact, as shown by elementary mathematical techniques, that F* is not novel; having been previously described under different names for more than a century in various scientific disciplines (weather forecasting, ecology, machine learning). Our preference is to call this measure the "critical success index" (CSI). It is a single measure which, in clinical medicine, combines information

about the classification performance with respect to both test and patient. It is easy to calculate, simple to use, and to interpret (higher values are better).

Mbizvo, G. K., & Larner, A. J. (2024). F*, an interpretable transformation of the F measure, equates to the critical success index. *Mathematics in Medical and Life Sciences*, *1*(1).

Reflect:

- What are some ideas you can use to help you establish common ground in your writing?

Bibliography

Annesley, T. M. (2011). Top ten tips for responding to reviewer and editor comments. *Clinical Chemistry, 57*(4), 551–554. https://doi.org/10.1373/clinchem.2011.162388.

Belcher, W. L. (2021). *Writing your journal article in Twelve Weeks: A guide to academic publishing success*. The University of Chicago Press.

Bondi, M. (2012). Voice in textbooks: Between exposition and argument. In K. Hyland & C. Sancho Guinda (Eds.), *Stance and voice in written academic genres* (pp. 102–115). Palgrave Macmillan.

Bradshaw, T. D., Turyanska, L., Thomas, N. R., Kuruppu, A., Breen, A., & Bouzijab, K. (2019, October 15-17). *Development of apoferritin formulations for antitumour drug delivery* [Conference presentation abstract]. First CA17140 conference of the European Cooperation in Science and Technology, Liga, Latvia. https://www.nano2clinic.eu/sites/default/files/downloads/CA1740COnference_BOA_0.pdf.

Bowden, D. (1999). *Mythology of voice*. Boynton/Cook.

Burford, J., & Henderson, E.F. (2022). *Making sense of academic conferences: Presenting, participating and organising* (Chapter 4). Routledge. https://doi.org/10.4324/9781003144885.

Sancho Guinda, C., & Hyland, K. (2012). Introduction: A context-sensitive approach to stance and voice. In K. Hyland & C. Sancho Guinda (Eds.), *Stance and voice in written academic genres* (pp. 1–11). Palgrave Macmillan. https://doi.org/10.1057/9781137030825_1

Silver, M. (2012). Voice and stance across disciplines in academic discourse. In K. Hyland & C. Sancho Guinda (Eds.), *Stance and voice in written academic genres* (pp. 50–69). Palgrave Macmillan.

Tse, P. (2012). Stance in academic bios. In K. Hyland & C. Sancho Guinda (Eds.), *Stance and voice in written academic genres* (pp. 70–85). Palgrave Macmillan.

Horn, S. A. (2016). The social and psychological costs of peer review: Stress and coping with manuscript rejection. Journal of Management Inquiry, 25(1), 11-26

Hyland, K. (2004). Disciplinary interactions: Metadiscourse in L2 postgraduate writing. *Journal of Second Language Writing, 13*, 133-151. https://doi.org/10.1016/j.jslw.2004.02.001.

Hyland, K. (2008). Disciplinary voices: Interactions in research writing. *English text construction*, *1*(1), 5-22.
Hyland, K., & Guinda, C. S. (Eds.). (2012). *Stance and voice in written academic genres* (pp. 134-150). Basingstoke: Palgrave Macmillan.
Lee, I. (2002). Teaching coherence to ESL students: A classroom inquiry. *Journal of second language writing, 11*(2), 135-159. https://doi.org/10.1016/S1060-3743(02)00065-6.
Lee, Y. Y., Lau, J. H., Vaingankar, J. A., Sambasivam, R., Shafie, S., Chua, B. Y., ... & Subramaniam, M. (2022). Sleep quality of Singapore residents: Findings from the 2016 Singapore mental health study. *Sleep Medicine: X, 4*, 100043. https://doi.org/10.1016/j.sleepx.2022.100043.
Major, P. (2017). *Is 'modern Confucianism' an oxymoron? Textual authority and the early new Confucian response to the modern tradition of anti-traditionalism.* ScholarBank@NUS Repository.
Matsuda, P. K. (2015, June 25). Feedback on a conference proposal. [Blog post]. Retrieved from http://dissoilogoi2.blogspot.com/2015/06/feedback-on-conferece-proposal.html.
Matsuda, P. K., & Tardy, C. M. (2007). Voice in academic writing: The rhetorical construction of author identity in blind manuscript review. *English for Specific Purposes (New York, N.Y.)*, *26*(2), 235–249.
Morton, J., & Storch, N. (2019). Developing an authorial voice in PhD multilingual student writing: The reader's perspective. *Journal of Second Language Writing*, *43*, 15–23.
Noyes, J., Hart, S., & Zanatto, D. (2023, May 22-25). *Role of time perception in automation* [Conference presentation abstract]. Seventeenth annual conference on psychology, Athens, Greece. https://www.atiner.gr/abstracts/2023ABST-PSY.pdf.
Pallas, C. L. (2022). *Doctoral student skills: Using your comparative advantage to succeed in grad school and prepare for the job market* (Chapter 7). Routledge. https://doi.org/10.4324/9781003262831.
Schultz, D.M. (2009). *Eloquent science: A practical guide to becoming a better writer, speaker, and atmospheric scientist* (Chapter 23). Boston, MA: American Meteorological Society. https://doi.org/10.1007/978-1-935704-03-4_23.
Tan, T. H. (2021). *How to read modern Chinese Literature in English? The women beside/s modernism.* Scholarbank@NUS Repository.
Tardy, C. (2016). 16 Voice and identity. *Handbook of second and foreign language writing*, *11*, 349-363.
Thomas, D. (2016). *The PhD writing handbook*. Palgrave.
Turabian, K. L., Colomb, G. G., Williams, J. M., Bizup, J., & Fitzgerald, W. T. (2023). *Student's Guide to Writing College Papers*. Chicago University Press.
Wallace, M., & Wray, A. (2021). *Critical reading and writing for postgraduates* (Chapter 19). Sage Publications Limited.
Wellington, J., Sikes, P., McCulloch, G., Hunt, C., & Bathmaker, A.-M. (2014). *Succeeding with your doctorate*. SAGE Publications Ltd.